

THO CHU, Viet Nam

WMO: 489160

Lat: 9.299N

Lon: 103.503E

Elev: 79

StdP: 14.65

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.9	75.7	65.3	93.8	77.9	67.3	100.7	78.7	24.3	80.6	21.9	80.1	5.7	270	0.358

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	92.5	82.2	91.0	81.7	89.7	81.2	83.2	90.2	82.5	89.2	81.9	88.0	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.7	87.6	80.7	159.8	86.9	80.1	156.6	86.0	47.2	90.6	46.5	89.6	45.8	88.6	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.3	13.4	72.4	96.0	2.3	2.0	70.7	97.4	69.3	98.6	68.0	99.8	66.3	101.3		
(5)			67.3	84.7	2.8	0.8	65.2	85.3	63.6	85.7	62.0	86.1	59.9	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	80.8	81.7	83.8	85.6	85.0	83.2	82.1	82.3	81.6	81.7	81.9	81.0
	DBStd	2.58	1.60	1.66	1.61	2.14	2.82	2.55	2.34	2.36	2.34	2.12	1.61	1.78
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11889	954	887	1047	1068	1086	997	997	1001	949	983	958	962
	CDD65	6414	489	467	582	618	621	547	532	536	499	518	508	497
	CDH74	67086	4366	4279	6132	7301	7458	6150	5617	5580	4978	5208	5245	4772
	CDH80	18971	671	893	1939	3042	3081	2037	1494	1456	1104	1190	1211	853
Wind	WSAvg	4.5	5.8	5.5	5.1	3.7	3.1	3.7	3.8	3.5	3.8	4.1	6.1	6.0
Precipitation	PrecAvg	89.8	1.4	0.8	1.7	3.4	7.2	10.8	12.7	10.5	14.3	13.3	9.3	4.4
	PrecMax	115.8	4.6	3.4	12.8	12.4	12.5	22.6	17.6	15.4	25.9	28.7	15.8	11.9
	PrecMin	69.3	0.0	0.0	0.0	0.0	0.2	2.8	4.4	5.0	7.5	4.8	3.6	0.7
	PrecStd	11.6	1.3	1.1	2.8	3.1	3.2	4.9	3.7	3.0	4.9	6.0	3.5	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.3	88.9	91.6	94.6	95.1	92.2	90.3	90.6	89.9	89.5	88.1	87.5
		MCWB	79.2	79.0	80.5	82.5	83.2	83.0	82.4	82.6	82.3	81.7	80.6	80.2
	2%	DB	85.6	87.2	89.8	92.5	92.7	90.0	88.4	88.6	88.0	87.5	86.7	85.9
		MCWB	77.9	78.3	79.6	81.8	82.5	82.2	81.4	81.7	81.3	80.8	80.0	79.1
	5%	DB	84.4	86.0	88.5	90.8	90.9	88.5	87.1	87.1	86.5	86.2	85.6	84.7
		MCWB	77.3	77.7	79.1	81.1	82.1	81.5	80.8	81.0	80.6	80.1	79.5	78.2
	10%	DB	83.3	84.6	87.2	89.4	89.1	86.9	85.6	85.7	85.0	84.9	84.5	83.6
		MCWB	76.7	77.0	78.6	80.6	81.5	80.8	80.2	80.3	79.9	79.5	78.8	77.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	80.0	81.6	83.2	84.2	83.9	83.2	83.3	82.8	82.1	81.7	81.0
		MCDB	86.1	87.1	89.8	92.2	92.5	90.3	88.9	89.0	88.6	88.1	86.9	86.3
	2%	WB	78.8	78.9	80.4	82.2	83.2	82.7	82.0	82.1	81.7	81.2	80.7	79.9
		MCDB	84.3	86.0	88.1	90.7	90.8	88.8	87.1	87.5	87.1	86.6	85.7	84.8
	5%	WB	78.0	78.1	79.7	81.5	82.4	81.8	81.1	81.3	80.8	80.5	79.9	79.0
		MCDB	83.1	84.6	86.9	89.4	89.6	87.2	85.8	86.2	85.8	85.4	84.5	83.6
	10%	WB	77.4	77.4	79.0	80.9	81.7	81.0	80.4	80.5	80.0	79.8	79.2	78.3
		MCDB	82.0	83.1	85.7	88.0	87.9	85.6	84.7	84.9	84.2	84.0	83.4	82.6
Mean Daily Temperature Range	5% DB	MDBR	6.4	8.1	9.4	9.9	8.6	7.3	6.9	7.3	6.8	6.8	6.3	5.8
		MCDBR	7.7	9.2	10.5	11.3	11.1	9.6	9.2	9.4	9.1	8.7	7.5	7.3
		MCWBR	3.6	4.0	3.9	3.9	4.2	4.4	4.6	4.7	4.3	4.2	3.8	3.6
	5% WB	MCDBR	7.4	9.0	10.4	11.0	10.3	9.0	8.6	9.1	8.8	8.4	7.4	7.2
		MCWBR	3.7	4.0	4.1	4.2	4.3	4.6	4.6	4.9	4.5	4.4	3.9	3.9
Clear-Sky Solar Irradiance	taub	0.508	0.511	0.519	0.480	0.439	0.445	0.447	0.444	0.436	0.459	0.471	0.486	
	taud	2.159	2.134	2.166	2.299	2.459	2.463	2.441	2.439	2.450	2.383	2.324	2.245	
	Ebn at Noon	256	259	259	266	272	267	267	272	277	271	265	259	
	Edh at Noon	48	51	50	43	36	35	36	37	37	39	41	43	
All-Sky Solar Radiation	RadAvg	1651	1874	1986	1998	1627	1447	1409	1468	1421	1466	1581	1568	
	RadStd	93	98	104	127	139	121	155	109	158	147	94	86	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.54	+0.81	N/A	N/A	+0.32	+0.32	N/A	N/A	N/A	N/A

Nomenclature: See separate page